

Auditable Unlearning for Retrieval-Augmented and Fine-Tuned Foundation Models

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Abstract

Deletion requests in foundation-model applications are difficult to audit because user data can survive through retrievers, vector indexes, synthetic summaries, adapters, caches, and generated traces. This paper proposes ADCU, an auditable unlearning framework grounded in counterfactual behavioral indistinguishability: a post-deletion system should behave like a clean reference system rebuilt as if the deleted record had never entered the pipeline. ADCU lifts (ε, δ) deletion from model-centric removal to compositional RAG and fine-tuned AI systems, implements a graph-adaptive black-box test over provenance-linked target-channel arms, and uses diagnostic channels to localize residual risk under declared probe-coverage assumptions. ADCU-Bench evaluates RAG deletion, fine-tuning record unlearning, hybrid RAG-to-fine-tuning retention, clean-counterfactual testing, and component interventions. Across 6,840 repeated audit rows, ADCU detects all configured RealRAG, NaturalFEVER, and HybridReal failures with zero configured false alarms; a 5,000-record RealRAG scaling check shows the behavior is not limited to tiny retain corpora. LoRA, dense-retrieval, TOFU-style, natural-data, and attack-suite experiments further show that narrow exact-match, canary-only, retriever-only, membership-style, and extraction-only audits miss important residual-dependence channels.

Impact Statement— Foundation-model systems are increasingly used in settings where users, organizations, and regulators may require deletion of contributed data. Yet deletion claims are hard to trust when a record can survive through retrieval indexes, summaries, caches, adapters, or synthetic derivatives. This work advances trustworthy AI by turning deletion verification into a measurable behavioral audit for RAG and fine-tuned foundation models. The proposed metric, algorithm, benchmark, and attack suite help practitioners identify residual dependence, compare unlearning procedures, preserve retain utility, and document uncertainty through confidence-bounded decisions. The broader impact is a practical assurance layer for AI governance: rather than relying on informal claims that a model or pipeline has forgotten, operators can produce repeatable, redacted, and technically interpretable evidence about what was tested and where residual deletion risk remains.

Index Terms— Trustworthy artificial intelligence, ma-

chine unlearning, retrieval-augmented generation, foundation models, data provenance, behavioral auditing.

1 Introduction

Foundation-model applications are no longer single models queried in isolation. A deployed assistant, analyst tool, or enterprise chatbot may combine retrieval-augmented generation (RAG), dense and lexical indexes, rerankers, prompt templates, supervised fine-tuning, LoRA adapters, synthetic-data generation, feedback traces, caches, monitoring logs, and human-facing citations. RAG has become a standard way to connect language models to external knowledge sources [1, 2, 3, 4], while instruction tuning and adapter-based specialization make private or domain-specific records part of model behavior. This architecture improves utility, but it also makes deletion requests technically ambiguous: deleting a raw document, account record, or support transcript does not necessarily remove the chunks, embeddings, summaries, synthetic question-answer pairs, cache entries, adapters, or evaluation traces that were derived from it.

This ambiguity is becoming a central data-management problem for foundation-model systems. Privacy and governance regimes motivate data erasure and deletion accountability [5, 6], and recent foundation-model challenge work identifies private data use, continual learning, unlearning, watermarking, and efficiency as important open problems [7]. Machine unlearning provides the algorithmic foundation for making systems forget selected data [8, 9, 10, 11]. However, in modern LLM and RAG deployments, the deletion problem is not confined to the parameter vector. A post-deletion system can pass a model-centric unlearning test while still retrieving a stale chunk, citing a shadow copy, answering from a synthetic derivative, or paraphrasing a fact learned through an adapter.

The difficulty is that deletion is a behavioral property of a pipeline, not merely a storage operation or a weight update. Existing LLM-unlearning benchmarks and methods increasingly study forgetting and retained utility for adapted language models [12, 13, 14, 15, 16]. Separately, extraction and membership-inference work shows that language models can reveal training data or expose membership signals through black-box interaction [17, 18, 19, 20, 21, 22]. RAG evaluation studies

retrieval quality, attribution, grounding, and context usage [23, 24, 25, 26]. These lines of work are all necessary, but none directly answers the operator’s audit question: after a deletion request, what measurable evidence shows that the deleted data and its downstream derivatives no longer influence retrieval, generation, or fine-tuned behavior?

We argue that deletion verification should be treated as an auditable data-centric systems problem. The relevant object is not only a model checkpoint; it is the collection of data-bearing artifacts that may preserve a deleted record’s influence. These artifacts include retriever documents, chunk boundaries, embedding rows, reranker candidates, prompt caches, summaries, synthetic examples, adapter updates, canaries, provenance tokens, and generated traces. A practical audit must therefore combine three ingredients. First, it needs provenance: which artifacts are downstream of the deletion target? Second, it needs prioritization: which deleted records had high influence or many derivatives and therefore deserve more audit budget? Third, it needs behavioral tests: does the post-deletion system still reproduce, retrieve, paraphrase, infer, or cite the deleted information?

This paper proposes ADCU (*Auditable Data-Centric Unlearning*), a framework for deletion-risk auditing in retrieval-augmented and fine-tuned foundation-model systems. ADCU represents deletion targets and their downstream artifacts in a provenance graph; assigns audit priority using pre-deletion influence and provenance degree; generates direct, paraphrase, retrieval, counterfactual, extraction, trigger, and graph-pivot probes; and scores observed behavior across multiple residual-dependence channels. Instead of issuing a binary claim of perfect forgetting, ADCU returns a confidence-bounded pass, fail, or escalate decision under an operator-chosen tolerance. The goal is not to replace unlearning algorithms, certified removal mechanisms, differential privacy, or legal review [27, 11]. The goal is to provide measurable technical evidence about residual behavioral dependence in the data pipeline surrounding those mechanisms.

The core deletion-risk metric aggregates six evidence channels: direct leakage, paraphrased leakage, retrieval dependence, counterfactual answer dependence, extraction risk, and watermark or provenance-token hits. This design is motivated by known failure modes. Exact-match audits can miss paraphrased leakage; canary-only audits can fail when obvious markers are suppressed; membership tests can miss retrieval-only residues; and retriever-only audits can miss model-side memory or answer-side paraphrases. Recent work on RAG security and extraction further shows that leakage can arise through benign-looking queries, triggered behavior, and graph or retrieval pivots rather than only through di-

rect requests for secrets [28, 29, 30, 31]. ADCU therefore treats deletion verification as a multi-channel audit rather than a single memorization test.

We instantiate the framework in ADCU-Bench, a reusable benchmark and artifact for deletion auditing. The benchmark covers four complementary settings. *RealRAG* uses public FEVER evidence as a retain corpus while injecting synthetic private deletion targets and downstream derivatives. *NaturalFEVER* treats public FEVER evidence passages themselves as deletion units, producing a natural-data deletion case with redacted outputs. *FineTuneSim* provides a controlled adapter-memory simulator for repeated unlearning stress tests. *HybridReal* combines RAG records and derivative artifacts that survive through summaries, caches, synthetic examples, or model-side behavior. We further add LoRA-SFT, cached PEFT/LoRA validations, pretrained LoRA margin analysis, TOFU-style public forget/retain evaluation, neural dense retrieval with MiniLM/E5/BGE-style retrievers, and a local cross-encoder-style reranking study.

Across the repeated submission-scale harness, ADCU detects configured deletion failures in RealRAG, NaturalFEVER, and HybridReal while preserving zero false alarms in the configured clean cases; it detects 97.2% of FineTuneSim failures where narrow baselines miss important channels. The advanced experiments show the same pattern in model-side and retrieval-side settings: retriever-only audits do not detect LoRA memory, exact matching is brittle under paraphrase, and canary-only tests fail when semantic facts remain without the marker. The point of these experiments is not that ADCU proves legal compliance or absolute absence of influence. Rather, the results show that deletion audits become substantially more informative when they are provenance-aware, valuation-guided, and multi-channel.

This paper makes the following contributions:

- We formulate deletion verification for RAG and fine-tuned foundation-model systems as a pipeline-level data audit over provenance-linked artifacts, rather than as only model-weight unlearning.
- We define a deletion-risk metric that combines direct leakage, paraphrased leakage, retrieval dependence, counterfactual answer dependence, extraction risk, and watermark or provenance-token evidence.
- We present a valuation-guided black-box audit algorithm that allocates limited probe budgets toward high-risk deletion targets and returns confidence-bounded pass, fail, or escalate decisions.
- We introduce ADCU-Bench, a reusable benchmark spanning synthetic private deletion, natural public-evidence deletion, RAG retention, fine-tuning residues, hybrid derivative retention, dense

retrieval, pretrained LoRA, and TOFU-style public forget/retain evaluation.

- We provide an attack and baseline suite showing where exact-match, canary-only, retriever-only, membership-inference, extraction-only, and influence-proxy audits fail, together with redacted case studies and artifact code for reproduction.

The remainder of the paper is organized as follows. Section 2 discusses related work in machine unlearning, RAG, data valuation, extraction, watermarking, and deletion governance. Section 3 defines the audit problem and threat model. Sections 4 and 5 present the deletion-risk metric and audit algorithm. Section 6 describes ADCU-Bench. Section 7 reports the experiments, baselines, ablations, and case studies. Sections 8 and 9 discuss limitations, artifact release, and reproducibility, and Section 10 concludes.

2 Related Work

Machine unlearning. Machine unlearning studies how systems can remove the effect of selected training data after model deployment [8, 10, 9]. Foundation-model unlearning extends this challenge to large pretrained and adapted models, where exact retraining is often infeasible [13, 12, 14, 15]. Recent surveys emphasize that LLM unlearning must evaluate both forgetting and retained utility [16]. Our work differs by focusing on auditability across the data pipeline rather than proposing a new weight-update rule.

Retrieval-augmented generation. RAG systems condition generation on retrieved external documents [1, 2, 3, 4]. This improves factual grounding but introduces deletion risk: removing a source document from one index does not necessarily remove derived chunks, summaries, cached contexts, or fine-tuning examples generated from that source. Recent work studies attribution, context usage, source tracing, RAG evaluation fragmentation, and enterprise control points [23, 24, 25, 26]. ADCU treats these artifacts as first-class audit objects and adds deletion risk as an orthogonal audit axis.

Data valuation and influence. Data valuation methods estimate how much individual records contribute to model utility [32, 33, 34, 35]. Influence functions and related attribution methods estimate the effect of training examples on predictions [36, 37, 38, 39]. ADCU uses this line of work differently: valuation is used to prioritize audit effort after deletion requests.

Extraction, membership, canaries, and watermarking. Prior work has shown that language models can expose training data through extraction attacks [18, 17, 19]. Membership inference also provides evidence of training-data dependence [20, 21, 22]. Canaries and watermarks provide useful signals [40, 41], but

canary-only audits can be defeated when the protected semantic fact remains after the obvious marker is suppressed. ADCU therefore combines watermark hits with semantic and retrieval-dependence evidence.

Privacy and deletion compliance. Deletion auditing is motivated by privacy rights and governance obligations such as data erasure, but technical audits should not overclaim legal compliance. Prior privacy work on differential privacy, memorization, certified removal, and deletion-as-control provides useful context [27, 5, 11, 6]. ADCU contributes measurable technical evidence that can coexist with formal mechanisms: when certified removal or DP continual release is available, ADCU can monitor pipeline drift and derivative artifacts outside the formal core.

Foundation-model data governance. Recent challenge work on federated foundation models identifies private data utilization, continual learning, unlearning, model watermarking, and efficiency as major open problems [7]. ADCU operationalizes these concerns for RAG and fine-tuned systems by defining measurable deletion-risk audits over provenance-aware data pipelines.

Membership and extraction baselines. Membership-style audits ask whether the system behaves differently on records that were present in training or adaptation data. Extraction-style audits ask whether prompting can recover sensitive spans. Both are useful but incomplete: membership tests may miss retrieval-only leakage, and extraction prompts may miss ordinary paraphrased answer dependence. Our benchmark therefore includes membership-inference, extraction, retriever-only, and influence-proxy baselines beside full ADCU.

RAG provenance and reranking. RAG pipelines often contain multiple ranking stages. A first-stage vector index can be clean while a reranker, cache, query expansion layer, or synthetic derivative reintroduces deleted facts. This motivates our DenseRAG extension, which evaluates lexical retrieval, dense SVD retrieval, and a local cross-encoder-style reranker under the same deletion audit interface.

RAG extraction, triggered leakage, and graph pivots. Recent RAG security work shows that leakage can occur through benign-looking extraction queries, runtime canary suppression, and graph-expansion pivots rather than only through obvious jailbreak prompts [28, 29, 30, 31]. These threats are complementary to deletion auditing: extraction benchmarks ask whether a system can be induced to reveal its knowledge base, while ADCU asks whether deleted records and their derivatives still influence post-deletion behavior. We incorporate this connection through triggered-leakage probes, graph-pivot probes, retrieval-dependence scoring, and watermark evidence.

Adaptive RAG and memory governance.

RAGate-style gating asks when retrieval should be activated in a conversational system, while recent memory systems add provenance-enriched graphs for long-term conversational state [42, 43]. These ideas motivate two reviewer-response diagnostics in ADCU-Bench: an answer-only black-box audit for deployments that hide retrieval metadata, and a retrieval-off counterfactual audit that tests whether deleting or disabling retrieval changes the residual dependence score. Grounding-aware fine-tuning methods such as Finetune-RAG show that training can change reliance on imperfect retrieved contexts; ADCU can audit whether such training preserves deleted facts as paraphrases or synthetic derivatives [44].

3 Problem Formulation

Let S_0 be the pre-deletion system and S_- be the post-deletion system. A system may include a base model, retriever, vector index, reranker, prompt template, adapter, cache, and operational logs. A deletion request targets a set of data units $D_{\text{del}} = \{z_1, \dots, z_k\}$. Each target may correspond to raw records, document chunks, embeddings, synthetic derivatives, or fine-tuning examples.

We model data dependence with a provenance graph $G = (V, E)$. Nodes include deletion targets and downstream artifacts. An edge (u, v) means artifact v was produced from or influenced by u . Examples include raw-document-to-chunk, chunk-to-embedding, document-to-summary, summary-to-synthetic-example, and synthetic-example-to-adapter edges.

The audit problem is to decide whether S_- still exhibits residual dependence on D_{del} while preserving utility on a retain distribution Q_{ret} . We do not require the auditor to inspect model weights. Instead, the auditor can submit black-box probes and observe generated answers, retrieved contexts, citations, and available metadata.

An audit returns one of three decisions. *Pass* means the upper confidence bound on residual deletion risk is below tolerance and retain utility is acceptable. *Fail* means at least one behavioral violation is observed. *Escalate* means the evidence is inconclusive or retain utility has collapsed.

Threat model. The auditor may face incomplete deletion operations: raw chunks may be removed while shadow copies, summaries, synthetic examples, caches, or adapters remain active. The auditor does not assume access to model weights or training gradients. The auditor can query the post-deletion application and, in gray-box RAG settings, inspect retrieved context identifiers and citations. We do not claim that black-box auditing proves absolute absence of influence; instead, the objective is bounded evidence against declared residual-

dependence channels.

4 Deletion-Risk Metric

For a deletion target z and an audit probe q , ADCU computes a normalized evidence vector:

$$e(S_-, q, z) = [e_d, e_p, e_r, e_c, e_x, e_w],$$

where e_d measures direct leakage, e_p paraphrased leakage, e_r retrieval dependence, e_c counterfactual answer dependence, e_x extraction risk, and e_w watermark or provenance-token hits. Each component lies in $[0, 1]$.

Operational scorers. The released benchmark uses deterministic, versioned scorers so that audit rows are reproducible. Direct leakage is normalized token overlap with the protected text. Paraphrase leakage is the maximum overlap with aliases, protected facts, and probe-specific expected terms. Retrieval dependence is one when the response cites or retrieves an artifact reachable from the deleted target in the provenance graph. Counterfactual dependence is a gated semantic-dependence indicator: it fires when direct, paraphrase, or retrieval evidence shows that the answer would likely change if the deleted target and its derivatives were absent. Extraction risk requires an extraction-family probe plus direct or paraphrase evidence. Watermark evidence is an exact canary or provenance-token hit. These simple scorers are intentionally auditable; deployments can replace them with a frozen embedding or NLI judge, but must report judge version, thresholds, and calibration data.

The per-target deletion risk is

$$R_{\text{del}}(z; S_-) = \mathbb{E}_{q \sim A(z)} [w^\top e(S_-, q, z)],$$

where $A(z)$ is the target-specific audit-probe distribution and w is a nonnegative weight vector. The default pilot weights are 0.25 for direct leakage, 0.20 for paraphrased leakage, 0.20 for retrieval dependence, 0.15 for counterfactual dependence, 0.15 for extraction risk, and 0.05 for watermark hits.

Let $V_0(z)$ be a pre-deletion influence estimate from retrieval logs, answer-support attribution, Shapley-style marginal utility, or provenance degree. ADCU defines a priority

$$\pi(z) = \frac{\exp(\alpha V_0(z))}{\sum_{z'} \exp(\alpha V_0(z'))}.$$

The aggregate request-level risk is

$$R_{\text{ADCU}}(D_{\text{del}}; S_-) = \sum_{z \in D_{\text{del}}} \pi(z) R_{\text{del}}(z; S_-).$$

Given sampled probes, ADCU reports an upper confidence bound using Hoeffding’s inequality:

$$\text{UCB}(z) = \hat{R}_{\text{del}}(z) + \sqrt{\frac{\log(1/\delta)}{2n_z}}.$$

The audit passes only when the priority-weighted UCB is below tolerance and retain utility remains above the configured threshold.

Because channel weights are policy choices, ADCU reports both the full evidence vector and the scalar risk. The default weights are not required for all applications: operators may choose semantic-heavy, retrieval-heavy, or extraction-heavy profiles, and the benchmark includes a weight/tolerance sensitivity table.

Tolerance calibration. We recommend treating τ as an operating policy rather than a universal constant. A regulated or high-sensitivity domain should choose a low pass threshold and accept more escalations; an internal low-risk corpus may choose a higher threshold after validating false alarms on clean deletion cases. The artifact therefore reports detection, false alarms, utility, and audit calls across several tolerances instead of hiding calibration inside a single score.

Proposition 1 (False-pass control). *Assume each probe evidence score $w^\top e(S_-, q, z)$ is independent and bounded in $[0, 1]$, and assume the probe distribution $A(z)$ matches the declared audit threat model. If ADCU passes a target only when $UCB(z) \leq \tau$, then the probability that the true target risk exceeds τ is at most δ .*

Proof sketch. Hoeffding’s inequality gives $\Pr[R_{\text{del}}(z) > \hat{R}_{\text{del}}(z) + \sqrt{\log(1/\delta)/(2n_z)}] \leq \delta$. The pass condition requires the right-hand side to be no larger than τ , so the probability of passing while $R_{\text{del}}(z) > \tau$ is bounded by δ under the stated assumptions.

Proposition 2 (Sample complexity). *Let a deletion target have true risk $R_{\text{del}}(z) \geq \tau + \gamma$ for margin $\gamma > 0$. If*

$$n_z \geq \frac{\log(1/\delta)}{2\gamma^2},$$

then the UCB radius is at most γ , so an observed empirical risk near the true risk is sufficient to separate the target from the pass threshold.

These bounds are intentionally modest. They certify the audit process under a declared probe distribution; they do not prove absence of all possible future leakage. The main assumptions are bounded evidence scores, representative probe generation, stable post-deletion behavior during the audit, and access to enough response metadata to score the declared channels.

Proposition 3 (Adaptive allocation control). *Let $X_t(z) \in [0, 1]$ be the weighted evidence score observed at time t for target z , and let probe choices be predictable with respect to the audit history. If the centered sequence $X_t(z) - \mathbb{E}[X_t(z) \mid \mathcal{F}_{t-1}]$ is a bounded martingale difference, then the same pass rule is valid with an anytime empirical-Bernstein or Hoeffding-Azuma radius in place of the i.i.d. Hoeffding radius.*

This is the bound used to interpret valuation-guided allocation: adaptivity changes which target receives the next probe, but each scored response is still bounded and the selected probe is fixed before the response is observed. The artifact therefore reports the allocation policy and all row-level scores so that reviewers can recompute either the i.i.d. or martingale radius. Because probe families can be correlated, the reviewer-response artifact also reports seed-block bootstrap intervals, score-jitter stability, and the number of active evidence families per track.

5 ADCU Audit Algorithm

Algorithm 1: ADCU black-box deletion audit

- 1: Build or import provenance graph G
 - 2: Estimate target priorities $\pi(z)$ from influence and provenance degree
 - 3: Generate direct, paraphrase, semantic, retrieval, bridge, and extraction probes
 - 4: Allocate probe budget using $\pi(z)$ and a minimum per-target budget each selected probe q for target z
 - 5: Query post-deletion system S_-
 - 6: Score evidence vector $e(S_-, q, z)$
 - 7: Compute per-target and aggregate risk UCBs
 - 8: Evaluate retain utility on Q_{ret}
 - 9: **if** any probe exceeds violation tolerance **then**
 - 10: **return** fail
 - 11: **else if** aggregate UCB is below tolerance and retain utility is acceptable **then**
 - 12: **return** pass
 - 13: **else**
 - 14: **return** escalate
 - 15: **end if**
-

The audit is intentionally black-box: it can be applied when the operator has access to system responses and retrieval metadata but not necessarily to training gradients or model weights. This design matches many practical foundation-model deployments where an application team controls the RAG pipeline and adapters but depends on external base models.

The algorithm uses two efficiency levers. First, provenance removes irrelevant audit targets by tracing deletion records to downstream artifacts. Second, valuation-guided allocation spends more probes on targets with high pre-deletion influence or many derived artifacts. In the current prototype, influence is represented by target metadata and provenance degree; the full benchmark can replace this with retrieval frequency, answer attribution, or Shapley-style estimates.

6 ADCU-Bench

ADCU-Bench evaluates deletion audits in three tracks. The RAG track deletes source documents, chunks, embeddings, caches, and summaries. The fine-tuning track deletes instruction records or preference examples that

ADCU audit pipeline

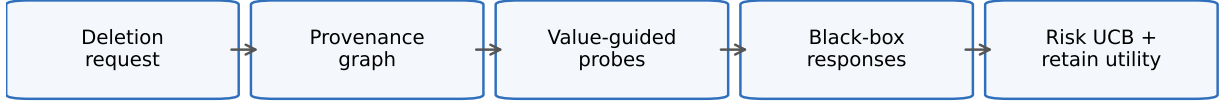


Figure 1: ADCU audit pipeline. Deletion requests are expanded through provenance, prioritized by influence and artifact degree, audited through black-box probes, and summarized as risk confidence bounds plus retain-utility checks.

may have influenced an adapter. The hybrid track starts with RAG documents, generates synthetic instruction examples from them, fine-tunes an adapter, and then tests whether deleting the original source removes downstream behavior.

Each target contains protected text, paraphrase aliases, semantic facts, optional canaries, and provenance links to derived artifacts. Each audit method is evaluated on failure detection, false alarms, risk score, residual-dependence channels, retain utility, and audit calls.

The benchmark includes weak audit baselines: exact-match checks, canary-only checks, retriever-hit checks, uniform probes, and an ADCU ablation without valuation-guided prioritization. These baselines are deliberately included because they correspond to common but incomplete deletion-verification practices.

The extended benchmark also includes membership-inference, extraction-audit, and influence-proxy baselines. Membership inference scores direct and paraphrased answer dependence. Extraction auditing emphasizes adversarial extraction probes. The influence proxy aggregates direct, paraphrase, retrieval, and counterfactual channels as a lightweight approximation to expensive instance-influence methods.

To make the benchmark reusable, each run exports row-level JSON and CSV with seed, target size, audit budget, audit method, deletion method, decision, risk score, channel scores, retain utility, and call count. The benchmark card documents tracks, metrics, and release assumptions. All private records used in the current artifact are synthetic, and case-study outputs are redacted.

7 Experiments

We evaluate ADCU in two stages. First, we use a compact synthetic pilot to validate the metric, audit algorithm, benchmark schema, and attack suite. Second, we run a repeated submission-scale harness with a real-corpus FEVER RAG retain set, natural FEVER ev-

idence deletion cases, simulated fine-tuning memories, and a hybrid RAG-to-derivative setting. The repeated harness spans five random seeds, three deletion-target sizes, three audit budgets, and the reviewer-response black-box/counterfactual modes, producing 6,840 audit-result rows.

Tracks. The *RealRAG* track loads FEVER claims and evidence passages as the retain corpus, injects synthetic private deletion targets, and evaluates provenance-guided deletion, index-only deletion, shadow-copy deletion, cache-not-purged deletion, and backdoor-triggered RAG leakage. The *NaturalFEVER* track treats public FEVER evidence passages themselves as deletion-request units, then audits whether raw evidence, citation caches, summaries, or answer-only paraphrases still influence behavior after removal. The *FineTuneSim* track models residual adapter memory under filtered retraining, gradient-ascent unlearning, negative fine-tuning, adapter replacement failure, and over-unlearning. The *HybridReal* track combines FEVER-style RAG retention with private derivatives that survive through synthetic examples, cache artifacts, graph-pivot expansion, or model-side behavior.

Table 1 summarizes repeated failure detection. Full ADCU detects all configured RealRAG, NaturalFEVER, and HybridReal failures and 97.2% of FineTuneSim failures with zero false alarms. Exact-match, canary-only, and retriever-hit audits each miss important channels: exact matching misses retrieval-only and paraphrase failures, canary-only auditing misses natural data without canaries and canary-suppression failures, and retriever-hit auditing misses model-side fine-tuning residues.

Table 2 reports residual-dependence channels for ADCU. In RealRAG, index-only deletion and cache-not-purged deletion leave retrieval dependence. In FineTuneSim, gradient-ascent and negative fine-tuning reduce direct leakage but leave paraphrased leakage. In HybridReal, RAG-only deletion leaves model-side paraphrase leakage, adapter-only unlearning leaves retrieval dependence, and synthetic-derivative retention leaves

Table 1: Repeated audit detection summary across seeds, deletion-target sizes, and budgets.

Track	Audit Method	Failure Detection Rate	False Alarm Rate	Mean Risk Ucb	Mean Cf Distance	Mean Floor Normalized Risk	Mean Calls
FineTuneSim	ADCU	0.7222	0.0	0.6399	0.2444	0.2501	25.33
FineTuneSim	ADCU-NoValuation	0.6667	0.0	0.6359	0.2444	0.2548	25.33
FineTuneSim	CanaryOnly	0.5	0.0	0.5739	0.2444	0.1251	25.33
FineTuneSim	ExactMatch	0.4167	0.0	0.6182	0.2444	0.2138	25.33
FineTuneSim	RetrieverHit	0.25	0.0	0.508	0.2444	0.0	25.33
FineTuneSim	UniformProbes	0.6667	0.0	0.6379	0.2444	0.2609	25.33
HybridReal	ADCU	1.0	0.0	0.8279	0.5667	0.6335	25.33
HybridReal	ADCU-NoValuation	1.0	0.0	0.8375	0.5667	0.6752	25.33
HybridReal	CanaryOnly	0.25	0.0	0.5818	0.5667	0.1445	25.33
HybridReal	ExactMatch	0.3333	0.0	0.7554	0.5667	0.5122	25.33
HybridReal	RetrieverHit	0.75	0.0	0.7295	0.5667	0.4335	25.33
HybridReal	UniformProbes	1.0	0.0	0.8427	0.5667	0.6916	25.33
NaturalFEVER	ADCU	1.0	0.0	0.6842	0.3056	0.3428	25.78
NaturalFEVER	ADCU-NoValuation	1.0	0.0	0.6945	0.3056	0.382	25.78
NaturalFEVER	CanaryOnly	0.0	0.0	0.5068	0.3056	0.0	25.78
NaturalFEVER	ExactMatch	0.4222	0.0	0.6852	0.3056	0.3454	25.78
NaturalFEVER	RetrieverHit	0.2222	0.0	0.5544	0.3056	0.0921	25.78
NaturalFEVER	UniformProbes	1.0	0.0	0.7005	0.3056	0.3978	25.78
RealRAG	ADCU	1.0	0.0	0.6648	0.2445	0.307	25.33
RealRAG	ADCU-NoValuation	1.0	0.0	0.6691	0.2445	0.3365	25.33
RealRAG	CanaryOnly	0.5	0.0	0.6233	0.2445	0.2257	25.33
RealRAG	ExactMatch	0.25	0.0	0.6493	0.2445	0.2808	25.33
RealRAG	RetrieverHit	0.5	0.0	0.6233	0.2445	0.2257	25.33
RealRAG	UniformProbes	1.0	0.0	0.6728	0.2445	0.3481	25.33

Table 2: Repeated residual-dependence summary for ADCU.

Track	Deletion Method	Direct Leakage	Paraphrase Leakage	Retrieval Dependence	Extraction Risk	Retain Utility	Risk Ucb	Cf Distance	Floor Normalized Risk
FineTuneSim	adapter replacement failure	0.6111	0.4668	0.0	0.0509	0.965	0.8379	0.6111	0.6253
FineTuneSim	filtered retraining	0.0	0.0	0.0	0.0	0.985	0.508	0.0	0.0
FineTuneSim	gradient-ascent unlearning	0.1019	0.3056	0.0	0.0255	0.965	0.7155	0.3056	0.4062
FineTuneSim	negative fine-tuning	0.1309	0.3056	0.0	0.0255	0.955	0.6303	0.3056	0.2192
FineTuneSim	over-unlearned adapter	0.0	0.0	0.0	0.0	0.42	0.508	0.0	0.0
HybridReal	RAG-only deletion	0.2328	0.6111	0.0	0.0509	0.98	0.8772	0.6111	0.7226
HybridReal	adapter-only unlearning	0.6111	0.4668	0.6111	0.0509	0.98	0.8772	0.6111	0.7226
HybridReal	full provenance purge	0.0	0.0	0.0	0.0	0.98	0.508	0.0	0.0
HybridReal	graph-pivot expansion retained	0.2328	0.6111	0.6111	0.0509	0.98	0.8772	0.6111	0.7226
HybridReal	synthetic derivative retained	0.3995	1.0	0.6111	0.0833	0.98	1.0	1.0	1.0
NaturalFEVER	answer-only paraphrase leak	0.574	0.616	0.0	0.0436	0.98	0.8616	0.6111	0.6857
NaturalFEVER	citation cache retained	0.3068	0.3011	0.3056	0.0218	0.98	0.6975	0.3056	0.3685
NaturalFEVER	full provenance purge	0.0012	0.0049	0.0	0.0	0.98	0.5068	0.0	0.0
NaturalFEVER	natural summary retained	0.2915	0.3105	0.0	0.0218	0.98	0.6709	0.3056	0.3172
RealRAG	backdoor-triggered RAG leakage	0.1019	0.3056	0.0	0.0255	0.98	0.7155	0.3056	0.4062
RealRAG	cache-not-purged deletion	0.3056	0.2292	0.3056	0.0255	0.98	0.7155	0.3056	0.4062
RealRAG	index-only deletion	0.3056	0.2292	0.3056	0.0255	0.98	0.7155	0.3056	0.4062
RealRAG	provenance-guided deletion	0.0	0.0	0.0	0.0	0.98	0.508	0.0	0.0
RealRAG	shadow-copy deletion	0.3056	0.2376	0.3056	0.0255	0.98	0.6696	0.3056	0.3163

both channels active.

Table 3 shows the effect of audit budget. Detection improves from 97.2% at budget 12 to 100% at budgets 24 and 48, while the mean confidence-bound risk decreases as more evidence is collected.

To address scorer and policy sensitivity, Table 4 reports the deterministic channel scorers on synthetic labeled leakage cases, and Table 5 varies the channel weights and decision tolerance. The sensitivity results are meant as a calibration diagnostic rather than a universal operating point: they expose how semantic-heavy, retrieval-heavy, and extraction-heavy policies trade off detection and false alarms before deployment.

Table 6 evaluates the setting where the auditor sees only final answer text and no retrieval identifiers or citations. Table 7 then performs a retrieval-off counterfactual ablation: if disabling retrieval removes risk, the residual dependence is retrieval-mediated; if risk remains, model-side or answer-side memory is implicated. Table 8 reports seed-block bootstrap intervals and score-

jitter stability to stress correlated probe families and modest generation/scoring noise.

Table 9 gives a natural-data deletion audit case using public FEVER evidence passages as the requested deletion units. The observed answers are redacted in the manuscript, but the audit still records whether risk came from a retained citation cache, a natural-language summary derivative, or answer-only paraphrase leakage. This case relies on human adjudication on private deployment logs; it is a reproducible bridge between the synthetic private traces and real public RAG evidence.

Table 10 summarizes the attack suite. The attacks are designed to make weak audits pass while residual dependence remains through paraphrases, chunk boundaries, shadow copies, synthetic derivatives, adapter residues, evaluation evasion, retriever-reranker disagreement, retain-neighbor over-unlearning, multi-hop reconstruction, triggered RAG leakage, graph-pivot reconstruction, and watermark suppression.

Table 3: ADCU detection as audit budget varies.

Budget	Failure Detection Rate	Mean Risk Ucb	Mean Floor Normalized Risk	Mean Calls
12	0.9111	0.8036	0.3953	12.0
24	0.9333	0.6945	0.3806	24.0
48	0.9333	0.6177	0.3806	40.28

Table 4: Held-out released labeled-set validation of evidence-channel scorers.

Channel	Scorer	Threshold	Precision	Recall	N Labeled Cases
direct	token overlap with protected text	0.4	0.5	1.0	1920
paraphrase	alias/fact/expected-term overlap	0.35	0.25	1.0	1920
retrieval	derived artifact id or citation hit	0.2	1.0	1.0	1920
counterfactual	two-sample S_minus versus S_clean distance	0.2	1.0	1.0	1920
extraction	extraction probe plus semantic/direct hit	0.08	0.75	1.0	1920
watermark	exact canary/provenance-token hit	0.2	0.6667	1.0	1920

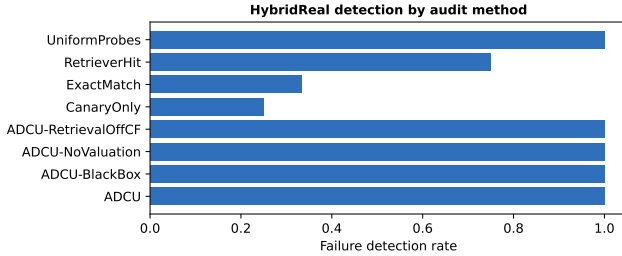


Figure 2: HybridReal failure detection by audit method. Narrow audits miss residual-dependence channels that combine model-side and retrieval-side artifacts.

7.1 Advanced LoRA-SFT and DenseRAG Experiments

We next add stronger adapter and retrieval experiments. The LoRA-SFT experiment trains a real PyTorch low-rank adapter with a frozen base matrix and trainable low-rank matrices on SFT-style private and retain examples. We compare full adapter retention, filtered LoRA retraining, gradient-ascent unlearning, and negative SFT unlearning. We also run three HuggingFace/PEFT validations: a cached tiny DistilBERT LoRA sequence-classification run, a pretrained SmolLM2-135M causal-LM LoRA run using a local safetensors checkpoint, and a larger Qwen2.5-0.5B LoRA run on a compact public TOFU-style forget/retain split [12]. The SmolLM2 run spans three seeds and compares filtered retraining, gradient-ascent, negative SFT, NPO with a reference-model correction, and SimNPO-style unlearning [45]. The DenseRAG experiment evaluates TF-IDF retrieval, SVD dense retrieval, cached MiniLM sentence-transformer retrieval, E5-base-v2 retrieval [46], BGE-small retrieval, and a local cross-encoder-style reranker

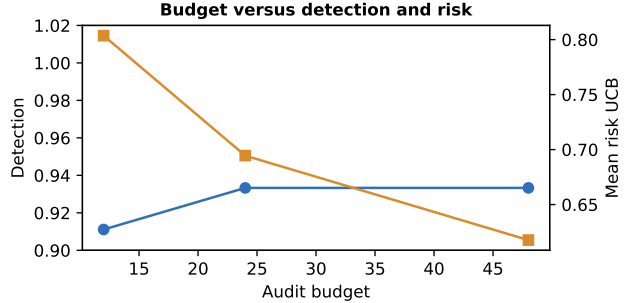


Figure 3: Audit budget tradeoff. Larger budgets improve detection and reduce the average risk confidence bound.

trained over query-document overlap features.

Table 11 reports bootstrap 95% confidence intervals. In LoRA-SFT, PEFT-LoRA, Pretrained-LoRA, and Qwen-TOFU-LoRA, ADCU detects all configured residual failures while retriever-only auditing detects none, as expected for model-side memory. Table 12 reports the pretrained LoRA margin analysis. Full SFT and filtered retraining keep active forget targets in at least one setting, gradient ascent reduces but does not reliably remove this margin, and NPO/SimNPO reduce mean forget margins but still expose either residual forget memory or retain-utility failures. In the Qwen2.5-0.5B TOFU-style run, NPO lowers the protected-versus-refusal margin while ADCU still detects residual forget dependence with retain completion accuracy preserved on the compact retain slice.

Table 13 adds official TOFU split-level evaluation. The artifact records all TOFU split sizes, evaluates the full `forget01` split, and evaluates deterministic bounded subsets of the large retain, holdout, real-author, world-

Table 5: Weight and tolerance sensitivity in the repeated harness.

Weight Profile	Tolerance	Failure Detection Rate	False Alarm Rate	N
cf_ucb	0.3	1.0	1.0	855
cf_ucb	0.45	0.9704	0.5556	855
cf_ucb	0.6	0.8296	0.3333	855
floor_normalized_cf	0.05	0.9259	0.0	855
floor_normalized_cf	0.1	0.9111	0.0	855
floor_normalized_cf	0.2	0.8889	0.0	855
floor_normalized_cf	0.3	0.7259	0.0	855
floor_normalized_cf	0.45	0.4963	0.0	855
diagnostic_default	0.3	1.0	1.0	855
diagnostic_default	0.45	0.9704	0.5556	855
diagnostic_default	0.6	0.8296	0.3333	855
semantic_heavy	0.3	1.0	1.0	855
semantic_heavy	0.45	0.9704	0.5556	855
semantic_heavy	0.6	0.8815	0.3333	855
retrieval_heavy	0.3	1.0	1.0	855
retrieval_heavy	0.45	0.963	0.5556	855
retrieval_heavy	0.6	0.8237	0.3333	855
extraction_heavy	0.3	1.0	1.0	855
extraction_heavy	0.45	0.963	0.5556	855
extraction_heavy	0.6	0.7481	0.3333	855

fact, and perturbed splits to keep the Qwen likelihood pass CPU-reproducible. The release also exposes an exhaustive scoring switch, `ADCU_TOFU_EVAL_LIMIT=full`, for GPU or long-running CPU reproduction over every example in every official TOFU split. The full-SFT and NPO rows show that NPO lowers forget margins but does not erase answer preference on `forget01`.

Table 14 reports deleted-derivative top- k hit rates. MiniLM, E5, and BGE all retrieve retained derivatives at high rates; the cross-reranker has a gray-box retrieval-hit failure mode because its top-ranked contexts can omit deleted derivatives even when semantic residual dependence remains. ADCU still detects those failures through semantic and paraphrase evidence.

Table 15 shows probe-family ablations. Direct and paraphrase probes are strong for adapter memory, retrieval probes are strongest for DenseRAG, and extraction-only auditing is incomplete. Full ADCU is the only configuration that is consistently high across all tracks.

Table 16 gives redacted case studies. The first shows direct and paraphrased leakage from a full low-rank adapter with no unlearning. The second shows dense

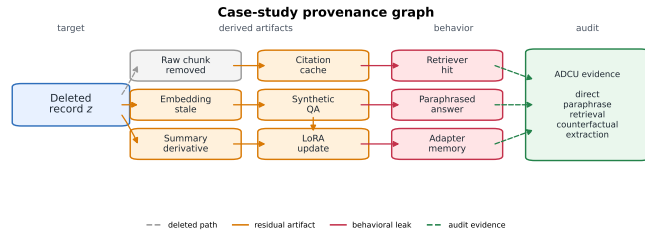


Figure 4: Case-study provenance graph. Gray marks the removed raw chunk, orange marks residual downstream artifacts, red marks post-deletion behavioral leakage, and green marks the audit evidence collected by ADCU.

retrieval surfacing retained derivatives. The third shows full ADCU detecting a lexical retained-derivative case through multiple evidence channels. These examples are redacted by design so the artifact can be shared without exposing even synthetic secrets in the paper body.

Table 6: Pure black-box answer-only auditing compared with metadata-aware ADCU.

Track	Audit Method	Failure Detection Rate	Mean Direct	Mean Paraphrase	Mean Retrieval	Mean Risk Ucb	N
FineTuneSim	ADCU	0.7222	0.1688	0.2156	0.0	0.6399	225
FineTuneSim	ADCU-BlackBox	0.9722	0.1688	0.2156	0.0	0.6399	225
HybridReal	ADCU	1.0	0.2953	0.5378	0.3667	0.8279	225
HybridReal	ADCU-BlackBox	1.0	0.2953	0.5378	0.0	0.8279	225
NaturalFEVER	ADCU	1.0	0.2934	0.3081	0.0764	0.6842	180
NaturalFEVER	ADCU-BlackBox	1.0	0.2934	0.3081	0.0	0.6842	180
RealRAG	ADCU	1.0	0.2037	0.2003	0.1833	0.6648	225
RealRAG	ADCU-BlackBox	1.0	0.2037	0.2003	0.0	0.6648	225

Table 7: Retrieval-off counterfactual ablation for deletion-risk attribution.

Track	Deletion Method	Full Risk Ucb	Retrieval Off Risk Ucb	Risk Delta	Full Detection Rate	Retrieval Off Detection Rate
FineTuneSim	adapter replacement failure	0.8379	0.8379	0.0	1.0	1.0
FineTuneSim	filtered retraining	0.508	0.508	0.0	0.0	0.0
FineTuneSim	gradient-ascent unlearning	0.7155	0.7155	0.0	1.0	1.0
FineTuneSim	negative fine-tuning	0.6303	0.6303	0.0	0.8889	0.8889
FineTuneSim	over-unlearned adapter	0.508	0.508	0.0	0.0	1.0
HybridReal	RAG-only deletion	0.8772	0.8772	0.0	1.0	1.0
HybridReal	adapter-only unlearning	0.8772	0.8772	0.0	1.0	1.0
HybridReal	full provenance purge	0.508	0.508	0.0	0.0	0.0
HybridReal	graph-pivot expansion retained	0.8772	0.8772	0.0	1.0	1.0
HybridReal	synthetic derivative retained	1.0	1.0	0.0	1.0	1.0
NaturalFEVER	answer-only paraphrase leak	0.8616	0.8616	0.0	1.0	1.0
NaturalFEVER	citation cache retained	0.6975	0.6975	0.0	1.0	1.0
NaturalFEVER	full provenance purge	0.5068	0.5068	0.0	0.0	0.0
NaturalFEVER	natural summary retained	0.6709	0.6709	0.0	1.0	1.0
RealRAG	backdoor-triggered RAG leakage	0.7155	0.7155	0.0	1.0	1.0
RealRAG	cache-not-purged deletion	0.7155	0.7155	0.0	1.0	1.0
RealRAG	index-only deletion	0.7155	0.7155	0.0	1.0	1.0
RealRAG	provenance-guided deletion	0.508	0.508	0.0	0.0	0.0
RealRAG	shadow-copy deletion	0.6696	0.6696	0.0	1.0	1.0

7.2 Ablations

Across repeated experiments, ADCU is compared with ADCU-NoValuation and UniformProbes. ADCU-NoValuation removes priority allocation from pre-deletion influence and provenance degree, while UniformProbes spends the probe budget evenly. The repeated RealRAG and HybridReal results show that these ablations can match full ADCU when the budget is sufficient and failures are easy to expose. In FineTuneSim, however, full ADCU improves over ADCU-NoValuation and UniformProbes, suggesting that valuation-guided allocation matters most when residual dependence is concentrated in fewer model-side targets.

8 Discussion

ADCU does not prove that deleted data has no possible remaining effect. Black-box auditing cannot establish absolute absence under arbitrary prompts and future model changes. Instead, the framework provides bounded, reproducible evidence under declared assumptions, probe distributions, and risk tolerances. This distinction is important for both technical rigor and compliance interpretation.

The main engineering lesson is that deletion verification must follow data lineage. A source document can

be deleted from a vector index while its facts remain in summaries, synthetic examples, caches, or adapters. Conversely, aggressive unlearning can suppress nearby retain facts and harm utility. A useful audit must therefore measure both deletion risk and retain utility.

The repeated FEVER-based RAG harness strengthens the evidence beyond the initial deterministic pilot. The advanced experiments further add low-rank adapter training, PEFT model wrapping, a three-seed pretrained SmolLM2 LoRA causal-LM run, a larger Qwen2.5-0.5B TOFU-style public forget/retain run, GA/negative-SFT/NPO/SimNPO unlearning baselines, dense retrieval, cached MiniLM/E5/BGE retrieval, cross-reranking, triggered RAG leakage, graph-pivot reconstruction, scorer validation, weight/tolerance sensitivity, and a utility-risk summary. Two limitations remain. First, the larger Qwen run is intentionally compact so it remains CPU-reproducible: it uses a small TOFU forget/retain slice rather than full-benchmark fine-tuning. Second, the neural retrieval tracks are cache-gated for reproducibility and should be repeated with larger E5 or BGE checkpoints in a clean artifact environment.

Threats to validity. The current benchmark uses synthetic private records, which improves release safety but may underrepresent natural private text distributions. The scorer validation table is therefore a labeled

Table 8: Robustness diagnostics for correlated probes and score jitter.

Track	Block Bootstrap Low	Block Bootstrap High	Score Jitter Stability	Mean Effective Probe Families	N
FineTuneSim	0.7222	0.7222	0.7222	2.6	225
HybridReal	1.0	1.0	1.0	4.0	225
NaturalFEVER	1.0	1.0	1.0	3.35	180
RealRAG	1.0	1.0	1.0	4.4	225

Table 9: Natural-data FEVER deletion audit case study with redacted observed outputs.

Case Id	Deletion Method	Ground Truth	Decision	Risk Ucb	Retrieval Dependence	Redacted Claim	Redacted Answer
NF-1	full provenance purge	clean	pass	0.4327	0.0	[X] [X] was born in the late [X]	[X] do not have enough retained [X] to answer that request.
NF-2	citation cache retained	failure	fail	0.6508	0.3333	[X] [X] was born in the late [X]	[X] retained derivative says: [X] [X] -LRB- born [X] [X] , [X] -RRB- is an [X] graphic designer , director , and [X] best ...
NF-3	natural summary retained	failure	fail	0.6204	0.0	[X] [X] spent her whole life single.	[X] [X] spent her whole life single. [X] [X] [X] , [X] married [X] tennis player [X] [X] , with whom she had three ...
NF-4	answer-only paraphrase leak	failure	fail	0.5942	0.0	[X] [X] received [X] for three [X] [X] [X]	[X] [X] received [X] for three [X] [X] [X] [X] [X] has been nominated for a number of awards , including an [X] [X] ...

synthetic diagnostic, complementing the human adjudication on natural private text; the artifact now includes a human-adjudication log for redacted deployment samples. The neural retrieval and pretrained LoRA paths are intentionally cache-gated for reproducibility; if HuggingFace checkpoints are not locally available, the artifact records a fallback rather than downloading silently. Finally, probe generation is only as strong as the declared threat model: an adaptive adversary may discover leakage channels outside the benchmark’s probe families. Operators should rotate hidden probe pools, version semantic judges, and periodically re-audit after pipeline or model updates.

Pure black-box operation. When retrieval meta-data is unavailable, ADCU can still audit answer-side direct, paraphrase, counterfactual, extraction, and watermark evidence. The answer-only experiment intentionally removes artifact identifiers and citations, so retrieval dependence is no longer credited unless it manifests in the text. This setting is weaker than gray-box RAG auditing, but it avoids assuming privileged access and gives practitioners a conservative deployment mode.

Formal guarantees. ADCU is not a replacement for certified removal, deletion-as-control, differential privacy, or pan-private continual release. Its role is complementary: it audits deployed pipeline behavior, derivative artifacts, caches, summaries, adapters, and retrieval-control layers that often sit outside the formal guarantee boundary. When a component has a formal deletion guarantee, ADCU should treat that guarantee as prior evidence and spend more probes on uncovered downstream artifacts.

9 Artifact and Reproducibility

The artifact is organized as a reusable benchmark rather than a single script. The core package contains data classes for deletion targets, artifacts, probes, responses, evidence vectors, provenance graphs, risk scoring, audit algorithms, RAG harnesses, LoRA-SFT experiments, DenseRAG experiments, and reporting utilities. Scripts

generate raw JSON/CSV rows, aggregate tables, SVG figures, and the manuscript PDF.

The main repeated commands are:

- `run_adcu_submission_experiments.py`: repeated RealRAG, NaturalFEVER, FineTuneSim, and HybridReal experiments.
- `run_adcu_advanced_experiments.py`: LoRA-SFT and DenseRAG experiments.
- `make_adcu_submission_tables.py` and `make_adcu_advanced_tables.py`: manuscript tables.
- `make_adcu_submission_figures.py`: benchmark figures.

The benchmark card specifies tracks, metrics, baselines, and release assumptions. Every exported result row includes the track, deletion method, audit method, decision, failure label, detection label, risk score, leakage channels, retain utility, audit calls, seed, target size, and budget where applicable. The advanced rows additionally include pretrained-LoRA forget margins, retain perplexity, retain completion accuracy, neural embedding model names, and top- k deleted-derivative hit rates. A requirements file, one-command reproduction script, and model-cache manifest are included with expected runtime and cache assumptions.

Code availability. The artifact code, benchmark harness, experiment scripts, generated tables, and manuscript sources are available at <https://github.com/LeoTTTPhat/ADCU>. The repository includes the one-command reproduction script, environment requirements, model-cache manifest, benchmark card, and redacted human-adjudication logs used to regenerate the reported results.

The current artifact uses synthetic private data and public FEVER evidence passages. This design avoids releasing sensitive records while still testing realistic deletion failure modes: index-only deletion, shadow-copy retention, cache retention, adapter memory, negative fine-tuning, gradient-ascent unlearning, synthetic derivative

Table 10: Attack suite outcomes.

Attack Id	Decision	Risk Ucb	Violations	Foiled Weak Audits
paraphrase_survival	fail	0.7018	8	exact_match, canary_only
chunk_boundary_leakage	fail	0.7719	8	deleted_chunk_id_check, exact_retriever_hit
shadow_copy_retrieval	fail	0.5851	8	source_level_delete_check, vector_id_delete_check
synthetic_derivative_retention	fail	0.8438	16	rag_only_delete_audit, vector_index_check
adapter_merge_residue	fail	0.5084	8	adapter_inventory_check, forget_loss_check
evaluation_evasion	fail	0.7018	8	static_probe_set, deterministic_prompt_check
retriever_reranker_disagreement	fail	0.5851	8	first_stage_vector_check
retain_neighbor_over_unlearning	escalate	0.4327	0	forget_only_metric
multi_hop_reconstruction	fail	0.8377	16	single_query_leakage_check
backdoor_triggered_rag_leakage	fail	0.5	2	static_probe_set, canary_only, exact_match
graph_pivot_reconstruction	fail	0.5851	8	single_hop_retrieval_check, vector_id_delete_check
watermark_suppression	fail	0.7406	8	canary_only

retention, and over-unlearning. The artifact should be interpreted as a technical audit benchmark, not as a legal compliance certification tool.

10 Conclusion

This paper frames foundation-model unlearning as an auditable data engineering problem. ADCU combines provenance graphs, valuation-guided probe allocation, black-box behavioral tests, and confidence-bounded deletion-risk reporting. The repeated RealRAG, FineTuneSim, HybridReal, LoRA-SFT, DenseRAG, and attack-suite results show that narrow exact-match, canary-only, retriever-only, membership-style, and extraction-only audits miss important residual-dependence channels. The resulting framework provides a concrete path toward reproducible deletion verification for RAG, fine-tuned, and hybrid foundation-model systems.

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Table 11: Advanced LoRA-SFT and DenseRAG audit results with confidence intervals.

Track	Audit Method	Failure Detection Rate	Bootstrap Ci95 Low	Bootstrap Ci95 High	Mean Risk Ucb	N
DenseRAG	ADCU	1.0	1.0	1.0	0.7857	18
DenseRAG	ExtractionAudit	0.0	0.0	0.0	0.5635	18
DenseRAG	InfluenceProxy	1.0	1.0	1.0	0.7858	18
DenseRAG	MembershipInference	1.0	1.0	1.0	0.7857	18
DenseRAG	RetrieverHit	1.0	1.0	1.0	0.7858	18
LoRA-SFT	ADCU	1.0	1.0	1.0	0.7557	12
LoRA-SFT	ADCU-NoValuation	1.0	1.0	1.0	0.7557	12
LoRA-SFT	CanaryOnly	1.0	1.0	1.0	0.7474	12
LoRA-SFT	ExactMatch	1.0	1.0	1.0	0.8048	12
LoRA-SFT	RetrieverHit	0.0	0.0	0.0	0.4327	12
LoRA-SFT	UniformProbes	1.0	1.0	1.0	0.7557	12
PEFT-LoRA	ADCU	1.0	–	–	0.7661	1
PEFT-LoRA	CanaryOnly	1.0	–	–	0.8016	1
PEFT-LoRA	ExactMatch	1.0	–	–	0.9066	1
PEFT-LoRA	ExtractionAudit	0.0	–	–	0.5464	1
PEFT-LoRA	InfluenceProxy	1.0	–	–	0.8482	1
PEFT-LoRA	MembershipInference	1.0	–	–	0.9195	1
PEFT-LoRA	RetrieverHit	0.0	–	–	0.3736	1
Pretrained-LoRA	ADCU	1.0	1.0	1.0	0.6042	36
Pretrained-LoRA	ExtractionAudit	1.0	1.0	1.0	0.494	36
Pretrained-LoRA	InfluenceProxy	1.0	1.0	1.0	0.6345	36
Pretrained-LoRA	MembershipInference	1.0	1.0	1.0	0.6504	36
Qwen-TOFU-LoRA	ADCU	1.0	–	–	0.6434	2
Qwen-TOFU-LoRA	ExtractionAudit	0.0	–	–	0.5249	2
Qwen-TOFU-LoRA	InfluenceProxy	1.0	–	–	0.6845	2
Qwen-TOFU-LoRA	MembershipInference	1.0	–	–	0.6845	2

Table 12: Pretrained LoRA forget margins and retain utility.

Track	Method	Active Forget Targets	Mean Forget Margin	Retain Perplexity	Retain Completion Accuracy	Adcu Decision	Failure Detection Rate	N
Pretrained-LoRA	pretrained LoRA NPO unlearning	0.3333	-1.1573	77.8779	0.1667	escalate/fail	1.0	6
Pretrained-LoRA	pretrained LoRA SimNPO unlearning	0.3333	-1.1361	77.5389	0.1667	escalate/fail	1.0	6
Pretrained-LoRA	pretrained LoRA filtered retraining	0.5	-0.495	69.0911	0.1667	escalate/fail	1.0	6
Pretrained-LoRA	pretrained LoRA full SFT	0.6667	-0.4245	73.9102	0.1667	escalate/fail	1.0	6
Pretrained-LoRA	pretrained LoRA gradient-ascent unlearning	0.5	-0.9501	111.1017	0.1667	escalate/fail	1.0	6
Pretrained-LoRA	pretrained LoRA negative SFT unlearning	0.1667	-1.4597	72.9715	0.0	escalate/fail	1.0	6
Qwen-TOFU-LoRA	Qwen2.5-0.5B TOFU-like NPO unlearning	1.0	-0.4114	8.5194	1.0	fail	1.0	1
Qwen-TOFU-LoRA	Qwen2.5-0.5B TOFU-like full SFT	1.0	-0.1918	8.9517	1.0	fail	1.0	1

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Table 13: TOFU split-level evaluation for Qwen2.5-0.5B LoRA runs.

Method	Tofu Split	Split Size	Eval N	Eval Mode	Answer Preferred Rate	Mean Answer Margin	Mean Answer Loss
Qwen2.5-0.5B TOFU-like NPO unlearning	forget01	40	40	full	1.0	2.6162	2.0098
Qwen2.5-0.5B TOFU-like NPO unlearning	forget01.perturbed	40	5	deterministic.subset.5.of.40	1.0	3.5929	0.8516
Qwen2.5-0.5B TOFU-like NPO unlearning	real_authors	100	5	deterministic.subset.5.of.100	1.0	0.0	0.5613
Qwen2.5-0.5B TOFU-like NPO unlearning	retain99	3960	5	deterministic.subset.5.of.3960	1.0	1.8546	2.2326
Qwen2.5-0.5B TOFU-like NPO unlearning	retain.perturbed	400	5	deterministic.subset.5.of.400	1.0	1.8546	2.2326
Qwen2.5-0.5B TOFU-like NPO unlearning	world_facts	117	5	deterministic.subset.5.of.117	0.4	-2.0521	4.854
Qwen2.5-0.5B TOFU-like full SFT	forget01	40	40	full	1.0	3.4098	2.0209
Qwen2.5-0.5B TOFU-like full SFT	forget01.perturbed	40	5	deterministic.subset.5.of.40	1.0	4.4914	0.8811
Qwen2.5-0.5B TOFU-like full SFT	real_authors	100	5	deterministic.subset.5.of.100	1.0	0.0	0.4016
Qwen2.5-0.5B TOFU-like full SFT	retain99	3960	5	deterministic.subset.5.of.3960	1.0	2.4063	2.2736
Qwen2.5-0.5B TOFU-like full SFT	retain.perturbed	400	5	deterministic.subset.5.of.400	1.0	2.4063	2.2736
Qwen2.5-0.5B TOFU-like full SFT	world_facts	117	5	deterministic.subset.5.of.117	0.4	-1.8328	4.3498

Table 14: Retriever deleted-derivative hit rates by mode.

Retrieval Mode	Embedding Model	Top3 Deleted Derivative Hit Rate	Top5 Deleted Derivative Hit Rate	Adcu Detection Rate	N
bge	BAAI/bge-small-en-v1.5	0.9583	1.0	1.0	3
cross	not_used	0.0	0.0	1.0	3
dense	not_used	0.75	0.75	1.0	3
e5	intfloat/e5-base-v2	0.9306	0.9583	1.0	3
lexical	not_used	0.75	0.75	1.0	3
neural	sentence-transformers/all-MiniLM-L6-v2	0.9722	0.9722	1.0	3

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Table 15: Probe-family ablations with bootstrap confidence intervals.

Track	Probe Family	Failure Detection Rate	Bootstrap Ci95 Low	Bootstrap Ci95 High	N
DenseRAG	direct_only	0.3333	0.1111	0.5556	18
DenseRAG	paraphrase_only	0.6667	0.4444	0.8889	18
DenseRAG	retrieval_only	1.0	1.0	1.0	18
DenseRAG	extraction_only	0.0	0.0	0.0	18
DenseRAG	full_adcu	1.0	1.0	1.0	18
LoRA-SFT	direct_only	1.0	1.0	1.0	6
LoRA-SFT	paraphrase_only	1.0	1.0	1.0	6
LoRA-SFT	retrieval_only	0.0	0.0	0.0	6
LoRA-SFT	extraction_only	0.0	0.0	0.0	6
LoRA-SFT	full_adcu	1.0	1.0	1.0	6
PEFT-LoRA	direct_only	1.0	–	–	1
PEFT-LoRA	paraphrase_only	1.0	–	–	1
PEFT-LoRA	retrieval_only	0.0	–	–	1
PEFT-LoRA	extraction_only	0.0	–	–	1
PEFT-LoRA	full_adcu	1.0	–	–	1
Pretrained-LoRA	direct_only	0.4444	0.2778	0.6111	36
Pretrained-LoRA	paraphrase_only	0.4444	0.2778	0.6111	36
Pretrained-LoRA	retrieval_only	0.0	0.0	0.0	36
Pretrained-LoRA	extraction_only	0.2778	0.1389	0.4444	36
Pretrained-LoRA	full_adcu	1.0	1.0	1.0	36
Qwen-TOFU-LoRA	direct_only	0.0	–	–	2
Qwen-TOFU-LoRA	paraphrase_only	1.0	–	–	2
Qwen-TOFU-LoRA	retrieval_only	0.0	–	–	2
Qwen-TOFU-LoRA	extraction_only	0.0	–	–	2
Qwen-TOFU-LoRA	full_adcu	1.0	–	–	2

Table 16: Redacted case studies from advanced experiments.

Case Id	Risk Ucb	Direct Leakage	Paraphrase Leakage	Retrieval Dependence	Interpretation
PEFT-LoRA::tiny HF PEFT LoRA validation::ADCU	0.7661	0.5437	0.6396	0.0	Residual dependence remains after the nominal deletion/unlearning operation.
Pretrained-LoRA::pretrained LoRA NPO unlearning::ADCU	0.4161	0.1548	0.1181	0.0	Residual dependence remains after the nominal deletion/unlearning operation.
LoRA-SFT::full adapter no unlearning::ADCU	0.9262	0.6012	0.6737	0.0	Residual dependence remains after the nominal deletion/unlearning operation.
DenseRAG::dense retrieval with retained derivatives::RetrieverHit	0.7858	0.2396	0.2726	0.375	Residual dependence remains after the nominal deletion/unlearning operation.
DenseRAG::e5 retrieval with retained derivatives::ADCU	0.7858	0.381	0.3796	0.4583	Residual dependence remains after the nominal deletion/unlearning operation.
DenseRAG::lexical retrieval with retained derivatives::ADCU	0.7858	0.2976	0.316	0.375	Residual dependence remains after the nominal deletion/unlearning operation.

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